Hyperbolas

ANSWER KEY

Section 1: Standard Equations of Hyperbolas

1 Find the vertices and orientation of $\frac{x^2}{16} - \frac{y^2}{9} = 1$.

a. **Horizontal, vertices at** $(\pm 4, 0)$ ✓

b. Vertical, vertices at $(0, \pm 3)$

c. Horizontal, vertices at $(\pm 5, 0)$

d. Vertical, vertices at $(0, \pm 4)$

Positive term is x^2, so the transverse axis is horizontal. $a^2 = 16 \Rightarrow a = 4$. Vertices are $(\pm 4, 0)$.

2 Find the vertices and orientation of $\frac{y^2}{25} - \frac{x^2}{144} = 1$.

a. **Vertical, vertices at** $(0, \pm 5)$ ✓

b. Horizontal, vertices at $(\pm 12, 0)$

c. Vertical, vertices at $(0, \pm 13)$

d. Horizontal, vertices at $(\pm 5, 0)$

Positive term is y^2, so the transverse axis is vertical. $a^2 = 25 \Rightarrow a = 5$. Vertices are $(0, \pm 5)$.

3 Find the length of the transverse axis and conjugate axis for $\frac{(x-1)^2}{36} - \frac{(y+2)^2}{64} = 1$.

a. **Transverse** $= 12$**, Conjugate** $= 16$ ✓

b. Transverse $= 6$, Conjugate $= 8$

c. Transverse $= 16$, Conjugate $= 12$

d. Transverse $= 72$, Conjugate $= 128$

$a^2 = 36 \Rightarrow a = 6 \Rightarrow 2a = 12$. $b^2 = 64 \Rightarrow b = 8 \Rightarrow 2b = 16$.

Section 2: Asymptotes of Hyperbolas

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4 State the equations of the asymptotes for the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$.

a. $y = \pm\frac{b}{a}x$ ✓

b. $y = \pm\frac{a}{b}x$

c. $y = \pm\frac{b^2}{a^2}x$

d. $y = \pm x$

For a horizontal hyperbola centered at the origin, the asymptotes pass through the center with slopes $\pm b/a$: $y = \pm\frac{b}{a}x$.

5 Find the asymptotes of $\frac{x^2}{9} - \frac{y^2}{16} = 1$.

a. $y = \pm\frac{4}{3}x$ ✓

b. $y = \pm\frac{3}{4}x$

c. $y = \pm\frac{16}{9}x$

d. $y = \pm\frac{5}{3}x$

$a = 3, b = 4$. Asymptotes: $y = \pm\frac{b}{a}x = \pm\frac{4}{3}x$.

6 Find the asymptotes of the vertical hyperbola $\frac{y^2}{4} - \frac{x^2}{9} = 1$.

a. $y = \pm\frac{2}{3}x$ ✓

b. $y = \pm\frac{3}{2}x$

c. $y = \pm\frac{4}{9}x$

d. $y = \pm 2x$

For a vertical hyperbola centered at $(0, 0)$, a^2 is under y^2 and b^2 is under x^2. Asymptotes are $y = \pm\frac{a}{b}x = \pm\frac{2}{3}x$.

Section 3: Foci and Eccentricity of Hyperbolas

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7 What is the relationship between a, b, and focal distance c for a hyperbola?

a. $c^2 = a^2 + b^2$ ✓

b. $c^2 = a^2 - b^2$

c. $a^2 = b^2 + c^2$

d. $c = a + b$

In a hyperbola, the foci lie farther from the center than the vertices, so $c^2 = a^2 + b^2$.

8 Find the foci of $\frac{x^2}{16} - \frac{y^2}{9} = 1$.

a. $(\pm 5, 0)$ ✓

b. $(0, \pm 5)$

c. $(\pm\sqrt{7}, 0)$

d. $(\pm 25, 0)$

$c^2 = 16 + 9 = 25 \Rightarrow c = 5$. Horizontal transverse axis: foci are $(\pm 5, 0)$.

9 Find the eccentricity $e = c/a$ of $\frac{x^2}{16} - \frac{y^2}{9} = 1$.

a. $5/4 = 1.25$ ✓

b. $4/5 = 0.8$

c. $3/4 = 0.75$

d. $\sqrt{7}/4$

$c = 5, a = 4$. Eccentricity $e = c/a = 5/4 = 1.25 > 1$.

Section 4: Identifying Conics from General Second-Degree Equations

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10 For $Ax^2 + Cy^2 + Dx + Ey + F = 0$ ($A, C \neq 0$), which condition identifies a hyperbola?

a. A **and** C **have opposite signs (**$AC < 0$**)** ✓

b. A and C have the same sign and $A = C$

c. A and C have the same sign and $A \neq C$

d. $A = 0$ or $C = 0$

If $AC < 0$, the quadratic terms have opposite signs, representing a hyperbola.

11 Identify the conic: $9x^2 - 4y^2 - 36x + 8y - 4 = 0$.

a. **Hyperbola** ✓

b. Ellipse

c. Parabola

d. Circle

$A = 9 > 0$ and $C = -4 < 0$. The signs are opposite, so this equation represents a hyperbola.

12 Convert $9x^2 - 4y^2 - 36x + 8y - 4 = 0$ to standard form.

a. $\frac{(x-2)^2}{4} - \frac{(y-1)^2}{9} = 1$ ✓

b. $\frac{(x-2)^2}{9} - \frac{(y-1)^2}{4} = 1$

c. $\frac{(y-1)^2}{9} - \frac{(x-2)^2}{4} = 1$

d. $(x-2)^2 - (y-1)^2 = 36$

$9(x^2 - 4x + 4) - 4(y^2 - 2y + 1) = 4 + 36 - 4 = 36$. Dividing by 36: $\frac{(x-2)^2}{4} - \frac{(y-1)^2}{9} = 1$.

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